Fiscal Year:	FY 2012	**Task Last Updated:**	FY 12/01/2011
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Project Title:	Radiation and Gliomagenesis: A Sensitive Model System to Evaluate the Tumorigenic Potential of HZE Particles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Space Biology Special Category:	None		
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Comments:	NOTE: Formerly at University of Texas Southwestern Medical Center at Dallas until fall 2019.		
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Radiobiology NNJ09ZSA001N
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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COI Name (Institution):	Bachoo, Robert (University of Texas Southwestern Medical Center) Boothman, David (University of Texas Southwestern Medical Center)		
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Task Description:	There is a paucity of systematic studies carried out with well-defined model systems to evaluate the tumorigenic potential of HZE particles that are presumed to pose a significant cancer risk to astronauts. We have shown that these particles induce DNA double-strand breaks (DSBs) that are repaired slowly and incompletely and trigger persistent DNA damage signaling events. In order to evaluate the carcinogenic consequences of such unrepaired DNA lesions, we propose to utilize genetically engineered astrocytes/neural stem cells (NSCs) as well as transgenic mouse models representing progressive steps in the development of Glioblastoma Multiforme (GBM). These lethal brain tumors are the third leading cause of cancer-related death among adults aged 30 to 50 years that are known to be induced by ionizing radiation. The past year has seen unprecedented advances in the genomic analyses of adult GBM tumors by the Cancer Genome Atlas Network, which reveal that these tumors have radically altered genomes with 5 key genetic changes dominating: loss of Ink4a, Arf, p53, or PTEN and amplification of EGFR (especially the constitutively active		

	EGFRvIII). We have developed astrocyte and NSC lines as well as mouse models with targeted deletions of these genes in logical combinations representing the progression of primary or secondary GBMs. This genetic progression model, though reductionist in approach, provides a realistic framework for evaluating and understanding how known genetic mutations may cooperate with charged particle radiation and contribute to the initiation and progression of high-grade gliomas. We hypothesize that poorly repaired DNA damage produced by charged particles will result in malignant progression towards brain cancer, and that this progression can be accelerated by predisposing oncogenic activations/tumor suppressor losses. More importantly, these "pre-initiated" murine astrocytes/NSCs will allow us to rapidly quantify the transforming potential of charged particles relative to gamma rays both in vitro and in vivo.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This grant seeks to understand cancer risks (especially the risk of brain cancers) from particle radiation in outer space. However, charged particles are also being used in the clinic for targeted tumor therapy. In addition to evaluating cancer risks to astronauts, our project will also contribute significantly to our understanding of cancer cell killing by charged particles in the clinic and can shed light on the risk of secondary cancers from particle therapy.
Task Progress:	Significance of funded research: Ions of high atomic number and energy (HZE particles) pose a significant cancer risk to astronauts on prolonged space missions. The decision to launch future long-duration space missions hinges, in part, upon the rapid estimation of cancer risks from these particles. Yet, there is a paucity of systematic studies carried out with well-defined model systems to evaluate the tumorigenic potential of HZE particles. In order to understand the carcinogenic consequences of heavy particle irradiation, we propose to utilize genetically engineered astrocytes/neural stem cells (NSCs) as well as transgenic mouse models representing progressive steps in the development of lethal brain tumors, Glioblastoma Multiforme (GBM). Our model systems provide a realistic framework for evaluating and understanding how known genetic mutations may cooperate with charged particle radiation and contribute to the initiation and progression of high-grade gliomas. With this backdrop, the specific aims of this project are: Aim 1: Evaluation of DNA damage responses to charged particles in neural stem cells (NSCs) and astrocytes in vitro and in vivo. Aim 2: Evaluation of the tumorigenic potential of charged particles relative to gamma rays in vitro and in vivo. Aim 3: Identification of the genetic and gene expression changes underlying particle-induced tumorigenesis. Brief summary of progress: In experiments carried out during the second year of the project, we have been able to evaluate the relative tumorigenic potential of heavy particle irradiation compared to gamma-rays using in vivo model systems. Using various mouse models, our results thus far indicate that HZE particles are more potently tumorigenic than gamma-rays. Detailed summary of progress: We have developed mouse models of gliomagenesis using genetically engineered mice in which deletion of the most critical, glioma-relevant tumor suppressor genes (Ink4a, Ink4b, Arf, P53 and PTEN) can be targeted to the mouse brain. Based on our published in vitro results, Nestin-Cre; Ink4ab-/-; Arf f/f mice were irradiated with a single dose of 1 Gy Fe ions (600 MeV) or 4 Gy gamma-rays. While gamma-induced damage is efficiently repaired by 1 day, unrepaired DNA lesions in the mouse brain are evident up to 1 month post-exposure following Fe irradiation. Most importantly, we find that combined loss of Ink4a, Ink4b and Arf cooperate with DNA damage by Fe ions resulting in a high incidence (25%) of malignant gliomas. These highly heterogeneous tumors arise with an average latency of 4.5 months and have all been classified as high grade (III and IV) glial tumors. Although tumors arising after gamma irradiation are also classified as high grade, they arise with a longer latency (5.6 months) and occur with significantly lower frequency (15%). Initial characterization of ex vivo cultures from these tumors indicates activation of the RTK-PI3K-Akt pathway by heterogeneous means such as cMET amplification, similar to that seen in human GBMs. Detailed analyses of these tumors using array CGH and microarray platforms is ongoing. Future plans: The long-term goal of this project would be to evaluate the carcinogenic potential of HZE particles using mouse models available, or currently being developed in our laboratory. These models consist of conditional knockout mice carrying floxed alleles of glioma- relevant tumor suppressors Ink4a, Ink4b, Arf, p53, PTEN or combinations thereof. Targeted deletion of these tumor suppressors is induced by expression of Cre recombinase driven by CystatinC and Glast promoters for astrocyte specific deletion, or Nestin and Mash1 promoters for deletion in neural stem cell (NSC) populations. Mice deficient in tumor suppressor genes p53 or Ink4a/Arf in either NSCs or astrocytes never spontaneously develop gliomas. However, we and others have shown that following addition of just one activating mutation (EGFR, PDGF, Met, cMyc, K-Ras, activated Akt, etc) or loss of an additional tumor suppressor (PTEN) these mice readily develop gliomas of various grades. We hypothesize that particle radiation will produce DNA damage leading to genomic changes that can cooperate with pre-existing mutations to trigger tumorigenesis in vivo. Thus far, we have shown that Nestin-Cre; Ink4ab-/-;Arf f/f transgenic mice are a sensitive and useful tool for evaluating the tumorigenic potential of HZE particles. Fe irradiation induces high grade gliomas at a much higher frequency compared to gamma-rays. We are currently developing Nestin-Cre; p53 f/f transgenic mouse models to examine another signature GBM mutation. Our goal is to compare the p53 tumor suppressor loss to 4ab and combined 4a/p53 backgrounds and select the most sensitive model that will allow us to look at tumorigenesis at radiation doses more typical of the space radiation environment (0.2-0.5 Gy) and further elucidate the spewise process of transformation. Future work will also examine how PTEN heterozygosity, which represents a progressive step towards the development of GBM, further cooperates with Ink4a, Ink4ab, Arf, and p53 loss to further decrease latency of tumor formation upon Fe irradiation. From the use of these mouse models we hope to gain basic insights into the cooperation between charged particles and pre-existing genomic lesions during the carcinogenic process. Most importantly, we hope to obtain a global and accurate picture of the specific genetic changes underlying transformation by HZE particles. The data obtained will also allow comparison of genomic and expression changes occurring in human gliomas that have been recently revealed by large scale genomic analyses. Hopefully, these comparisons will allow us to understand whether HZE-induced gliomagenesis is fundamentally similar to that occurring spontaneously, both in mouse models and in human patients.

Bibliography Type:	Description: (Last Updated: 12/09/2025)
Articles in Peer-reviewed Journals	Mukherjee B, Burma S. "DNA double-strand breaks make bedfellows of ATM and AKT." Cell Cycle. 2011 Oct 1;10(19):3230. PubMed PMID: 21946524 ; http://dx.doi.org/10.4161/cc.10.19.17080 ; Free access PDF, accessed 12/02/2011: http://www.landesbioscience.com/journals/cc/CC10-19-News%20and%20Views-Mukherjee.pdf , Oct-2011